

Student Number: _____



SCEGGS DARLINGHURST

TRIAL EXAMINATION

Mathematics

Year 12

2/3 Unit

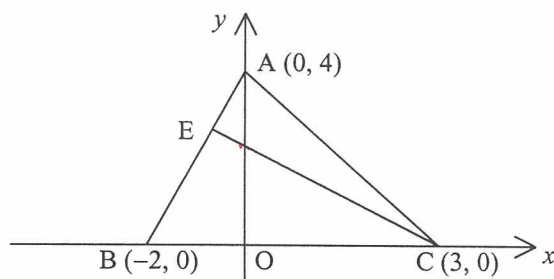
TIME ALLOWED: 3 hours (plus 5 minutes reading time)

DIRECTIONS TO CANDIDATES:

- Attempt all ten questions. All questions are of equal value.
- Ensure that your student number is on this paper.
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Approved calculators should be used. Mathematical templates and geometrical instruments may be used.
- Begin each question on a new page - write your number at the top of each page.
- The table of standard integrals is printed on the last page.

Question 1. (12 Marks)**Marks**

- a) Find, to two significant figures, the value of $\sqrt{\frac{3.5^2 - 4}{4.6 - 2.5}}$. 2
- b) Factorise $4x^2 - 36$. 2
- c) Simplify: $3 - 2(3x + 4) + 13x$. 2
- d) Solve $15x^2 = 10x$. 2
- e) Show the solution to $|x - 4| < 5$ on a number line. 2
- f) Show that the reciprocal of $(\sqrt{3} - \sqrt{2})$ is $(\sqrt{3} + \sqrt{2})$ 2

Question 2. (12 Marks) *Begin a new page.*

In the figure shown, ABC is a triangle with vertices A (0, 4), B (−2, 0) and C (3, 0). AO and CE are altitudes of the triangle (that is, AO is perpendicular to BC and EC is perpendicular to AB).

Copy the diagram onto your writing paper.

- a) State the equation of the altitude AO. 1
- b) Show that the equation of EC is $x + 2y - 3 = 0$. 2
- c) Find the co-ordinates of H, the point of intersection of EC and AO. 1
- * d) Find the ratio, OH : AH. NB. should have 3:5 to get 1 1
- e) Explain why the equation of the line AB is $y = 2x + 4$. 2
- f) Find the length EC, that is, the perpendicular from C to E. 2
- g) Find the area of the triangle ABC using AB as the base and EC as the height. 3
Check your answer by finding the same area another way.

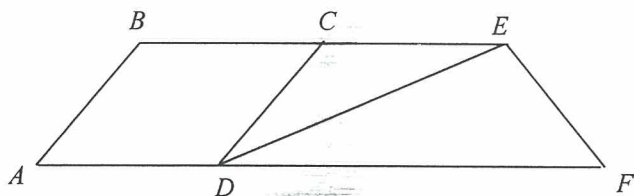
Question 3 begins on the next page.

Question 3. (12 Marks) Begin a new page.**Marks**

- a) Differentiate: 3
- i) $\tan(5x - 3)$
- ii) $\frac{2}{3x^2}$
- b) i) Rewrite $\log_e\left(\frac{2x-1}{4x+1}\right)$ using the Logarithm laws. 3
- ii) Hence differentiate $\log_e\left(\frac{2x-1}{4x+1}\right)$
- c) Find 4
- i) $\int (3x + 2)^4 dx$
- ii) $\int_0^1 3e^{2x} dx$
- d) *I didn't take 1 mark off if had - or time*
 A body is increasing its velocity at a decreasing rate. Draw a sketch of the velocity-time graph for this body. 2

Question 4. (12 marks) Begin a new page.

- a) 'When tossing three coins there are four possible results. They are: 3 tails, 2 tails and 1 head, 2 heads and 1 tail, 3 heads. Thus the probability of obtaining 3 tails when tossing three coins is $\frac{1}{4}$.' 2
- Comment on the validity of this statement, justifying your response.
- b) In the figure below $ABCD$ is a rhombus. E is a point on BC produced such that C is the midpoint of BE . F is a point on AD produced such that $\angle DEF = 90^\circ$. Also $\angle BAD = 72^\circ$. 4



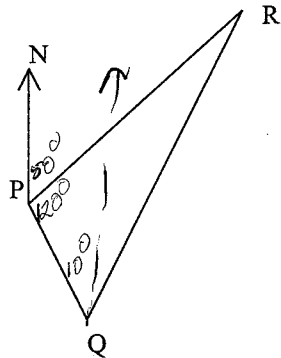
- i) Copy the diagram and clearly show this information.
- ii) Find, giving reasons, the size of $\angle CDE$.
- iii) Find, giving reasons, the size of $\angle DFE$.

Question 4 continues on the next page.

Question 4. (continued)**Marks**

- c) A ship leaves a port P, sailing on a bearing of 170° , and reaches a port Q, 50 kilometres from P. The ship's captain then realises that they should have sailed from P on a bearing of 50° to reach R, 170 km from P. The diagram is shown below.

6



- Copy the diagram and show on it the lengths of PR and PQ, as well as the size of the angle RPQ.
- Calculate the distance from Q to R, giving your answer to the nearest kilometre.
- Find $\angle PQR$ to the nearest degree, and hence the bearing on which the ship will need to sail so that it reaches R from Q.

Question 5. (12 Marks) Begin a new page.

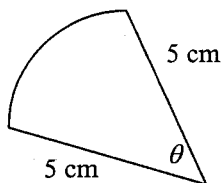
- a) A parabola has equation $(x - 4)^2 = 8(y + 3)$.

4

- Find the co-ordinates of its focus.
- Find the equation of its directrix.
- Describe the locus in words.

- b) The perimeter of the sector shown below is 18 cm.

4



- Find the exact value of the angle θ in radians.
 - Hence find the area of this sector.
- c) Consider the equation $x^2 + 6x + c = 0$.

4

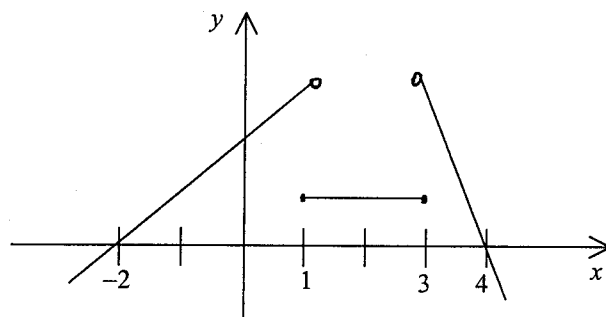
- Find the range of values of c for which there are no real solutions to this equation.
- Explain how you can use your answer in c) part i) to help you determine the values of x for which $x^2 + 6x + 20 > 0$, and what this tells you about the graph of $y = x^2 + 6x + 20$. (You do not need to draw this graph).

Question 6 begins on the next page.

Question 6. (12 Marks) Begin a new page.**Marks**

- a) The graph of
- $y = f(x)$
- is below.

3

Draw a sketch of $y = f'(x)$ for this function.

- b) For the curve
- $y = x^3 - 3x^2 + 1$
- ;

9

- Find any stationary points and determine their nature.
- Find any points of inflexion.
- Find the values of x for which the curve is concave down.
- Sketch the curve for $-1 \leq x \leq 4$.

Question 7. (12 marks) Begin a new page.

- a) Solve the inequality
- $-6 \leq 2x \leq 5 - 2x$
- .

2

- b) A box contains 12 light bulbs that look the same, but two of them are defective. Two light bulbs are drawn at random without replacement.

4

- What is the probability that the first light bulb drawn is defective?
- What is the probability that, of the two light bulbs drawn, at least one is defective?

- c) Explain how you can tell that a function has a horizontal point of inflexion. Give an example of the equation for such a function.

2

- d) Given that
- $\frac{dy}{dx} = \frac{x}{x^2 - 4}$
- ;

4

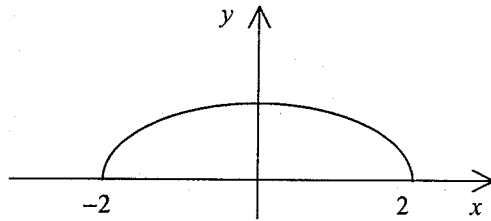
- Find y in terms of x , given that $y = 0$ when $x = 3$.
- State the set of x values for which y exists.

Question 8 begins on the next page.

Question 8. (12 Marks) Begin a new page**Marks**

- a) The diagram below shows the graph of the curve $y = \frac{1}{2}\sqrt{4-x^2}$.

6



- Use Simpson's Rule with five function values to estimate the area enclosed by the curve and the x axis.
 - Use integration to find the exact value of the volume of the solid of revolution formed when the region enclosed by this curve and the x axis is rotated about the x axis.
- b) The amount in grams of a radioactive substance present after t years is given by $A = 80e^{-0.025t}$.
- Find the amount of radioactive substance present after 10 years.
 - What is the half-life of this radioactive substance?
 - At what rate is this substance decaying after 8 years?

6

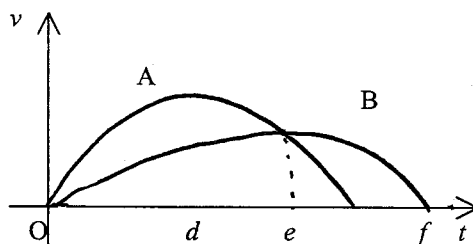
Question 9. (12 Marks) Begin a new page

- a)
 - Draw the graphs of $y = 2\sin x$ and $y = \tan x$, for $0 \leq x \leq 2\pi$.
 - Use your graphs to determine the number of solutions to $2\sin x = \tan x$ within this domain.
 - From the equation $2\sin x = \tan x$, show that $\sin x(2\cos x - 1) = 0$.
 - Find all values of x , where $0 \leq x \leq 2\pi$, that are solutions to the equation $2\sin x = \tan x$. (Give values in exact form.)

6

- b) Two particles A and B are moving in a straight line. The graphs below show the velocities (in m/s) of each particle at time t seconds.

6

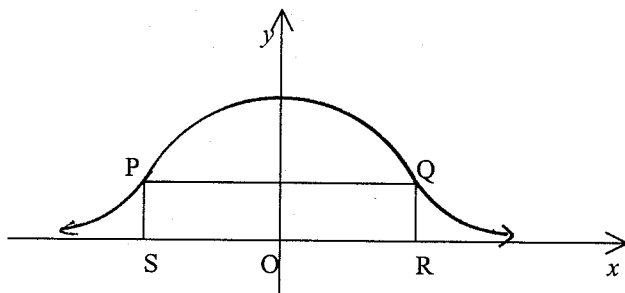


- Which particle shows the greater acceleration in the first d seconds? Why?
- Describe the motion of each particle at time e seconds.
- Draw a graph showing the displacement of particle B, as a function of time t , from $t = 0$ to $t = f$.

Question 10 begins on the next page.

Question 10. (12 Marks) *Begin a new page.***Marks**

- a) The diagram shows a rectangle PQRS, where P and Q are on the curve $y = e^{-x^2}$ and R and S are on the x axis. The point O is the origin and $OS = OR$. 5



- i) Let the length $OR = x$. Show that the area of the rectangle PQRS is represented by $A = 2xe^{-x^2}$.
 - ii) Find the value of x , in exact form, for which PQRS has maximum area.
- b) If $f(x) = 2 - \log_e x$, 7
- i) Explain why the graph of the curve $y = f(x)$ is always decreasing.
 - ii) Find the x intercept of $y = f(x)$.
 - iii) Sketch the graph of $y = f(x)$.
 - iv) Write down the integral that enables us to find the area enclosed by the curve $y = 2 - \log_e x$ the line $y = 2$ and the x -axis. (You do not have to evaluate this integral).

End of Examination.

1) a) $\sqrt{\frac{3.5^2 - 4}{4.6 - 2.5}} \div 1.98206 \dots$
 $= 2.0$ (to 2 sig figs)

b) $4x^2 - 36 = 4(x^2 - 9)$
 $= 4(x-3)(x+3)$

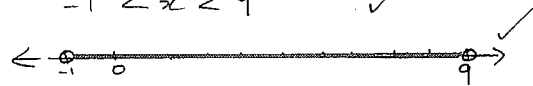
c) $3 - 2(3x+4) + 13x$
 $= 3 - 6x - 8 + 13x$
 $= 7x - 5$

d) $15x^2 = 10x$
 $15x^2 - 10x = 0$
 $5x(3x-2) = 0$

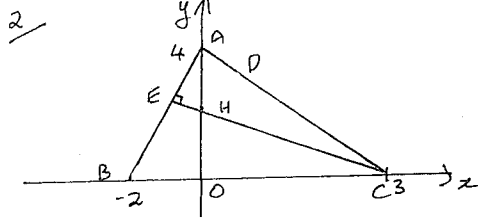
$x = 0, 3x - 2 = 0$
 $3x = 2$
 $x = \frac{2}{3}$

$\therefore$ Solutions are $x = 0, x = \frac{2}{3}$

e) $|x - 4| < 5$
 $-5 < x - 4 < 5$
 $-1 < x < 9$



f) $\frac{1}{\sqrt{3}-\sqrt{2}} = \frac{1}{\sqrt{3}-\sqrt{2}} \times \frac{\sqrt{3}+\sqrt{2}}{\sqrt{3}+\sqrt{2}}$
 $= \frac{\sqrt{3}+\sqrt{2}}{3-2}$
 $= \sqrt{3}+\sqrt{2}$



a) $AO: x = 0$
b) $m_{AB} = \frac{0-4}{-2-0} = 2$
 $\therefore m_{EC} = -\frac{1}{2}$
 $\therefore y - 0 = -\frac{1}{2}(x-3)$
 $2y = -x + 3$
 $x + 2y - 3 = 0$

c) $x = 0, 3x - 4y + 6 = 0$
 $-4y + 6 = 0$
 $-4y = -6$
 $y = \frac{6}{4} = \frac{3}{2}$
 $\therefore H = (0, \frac{3}{2})$
d) $OH: AH = \frac{3}{2} : (4 - \frac{3}{2})$
 $= \frac{3}{2} : \frac{5}{2}$
 $= 3 : 5$

e) $m_{AB} = \frac{4}{2} = 2 \rightarrow y = mx + b$
 $\therefore y = 2x + b$
 $b = 4$ (y-intercept)
 $\therefore y = 2x + 4$

f) $(3, 0) \quad 2x - y + 4 = 0$
 $EC = \frac{|2 \times 3 - 1 \times 0 + 4|}{\sqrt{2^2 + 1^2}}$
 $= \frac{|6 + 4|}{\sqrt{5}}$
 $= \frac{10}{\sqrt{5}} = \frac{10\sqrt{5}}{5} = 2\sqrt{5}$

g) $AB = \sqrt{4^2 + 2^2}$
 $PS = 3 \quad = \sqrt{20} = 2\sqrt{5}$

2g) ctd Area $\triangle ABC = \frac{1}{2} \times AB \times EC$
 $= \frac{1}{2} \times 2\sqrt{5} \times 2\sqrt{5}$
 $= 10 \text{ units}^2$

Area $\triangle ABC = \frac{1}{2} \times BC \times AO$
 $= \frac{1}{2} \times 5 \times 4$
 $= 10 \text{ units}^2$

3 a) i) $\frac{d}{dx} \tan(5x-3) = 5 \sec^2(5x-3)$

ii) $\frac{d}{dx} \left(\frac{2}{3x^2} \right) = \frac{d}{dx} \left(\frac{2}{3} x^{-2} \right)$
 $= \frac{2}{3} x^{-3}$
 $= -\frac{4}{3x^3}$

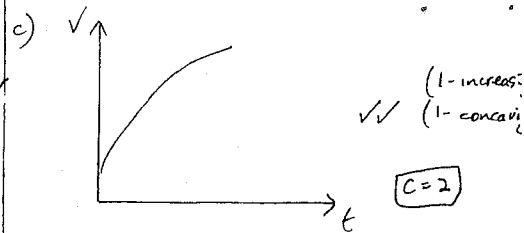
b) i) $\log_e \frac{(2x-1)}{4x+1} = \log_e(2x-1) - \log_e(4x+1)$

ii) $\frac{d}{dx} \log_e \frac{(2x-1)}{(4x+1)} =$

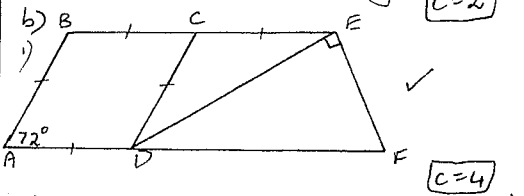
$\frac{d}{dx} \{ \log_e(2x-1) - \log_e(4x+1) \}$
 $= \frac{2}{2x-1} - \frac{4}{4x+1} = \frac{6}{(2x-1)(4x+1)}$

c) i) $\int (3x+2)^4 dx = \frac{(3x+2)^5}{5 \times 3} + C$
 $= \frac{1}{15} (3x+2)^5 + C$

ii) $\int_0^1 3e^{2x} dx = 3 \left[\frac{e^{2x}}{2} \right]_0^1$
 $= 3 \left(\frac{e^2}{2} - \frac{e^0}{2} \right)$

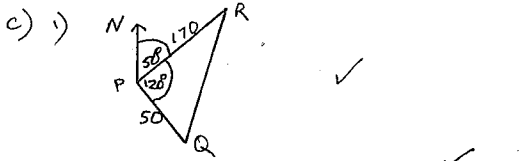


4 a) This statement is not valid since the probability of obtaining each of these outcomes is not the same - the outcomes are not equally likely. $C=2$



ii) $\angle BCD = 72^\circ$ (opp $\angle$ s in rhombus are equal)
 $\angle DCE = 180^\circ - 72^\circ$ (supplementary $\angle$ s)
 $= 108^\circ$
 $\therefore \angle CDE = \frac{1}{2}(180^\circ - 108^\circ)$ ($\angle$ s in isosceles triangle CED)
 $= 36^\circ$

iii) $\angle DFE = 90^\circ - 36^\circ$ ($\angle$ sum of right angled $\triangle DEF$)
 $= 54^\circ$



ii) $QR^2 = 50^2 + 170^2 - 2 \times 50 \times 170 \cos 120^\circ$
 $= 2500 + 28900 - 17000 \times -\frac{1}{2}$
 $= 31400 + 8500$
 $= 39900$
 $QR \approx 199.7498 \dots$
 $= 200 \text{ km}$ (to nearest km)

1) $\frac{170}{\sin \angle PQR} = \frac{200}{\sin 120^\circ}$ ✓
 $\sin \angle PQR = \frac{170 \sin 120^\circ}{200}$
 $= \frac{17 \times \sqrt{3}}{20 \times 2}$
 $= 0.7361 \dots$ ✓
 $\angle PQR = 47^\circ 24'$
 $= 47^\circ$ (to nearest degree) ✓



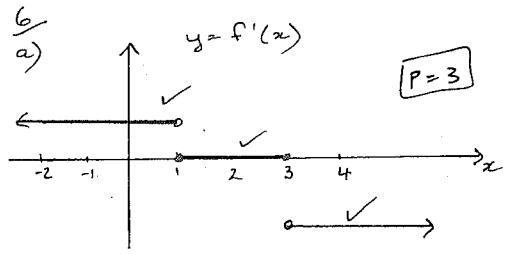
2) a) $(x-4)^2 = 8(y+3)$
 Vertex = $(4, -3)$
 $4a = 8$
 $a = 2$ ✓
 Focus = $(4, -1)$ ✓
 ii) directrix $y = -5$ ✓
 iii) This is the locus of all points that are the same distance from $(4, -1)$ as they are from $y = -5$ ✓

b) 1) $l = 10 = 50$ ✓
 Perimeter
 $= 5 + 5 + 50$
 $\therefore 10 + 50 = 18$ ✓
 $50 = 8$
 $\theta = \frac{8}{5}^\circ$ ✓

1) Area = $\frac{1}{2} 5^2 \times \frac{8}{5}$ ✓
 $= \frac{1}{2} \times 25 \times \frac{8}{5}$
 $= 20 \text{ cm}^2$

c) $x^2 + bx + c = 0$
 i) $\Delta = b^2 - 4ac$
 $= 36 - 4 \times 1 \times c$
 $= 36 - 4c$
 No real solns, $\Delta < 0$ ✓
 $\therefore 36 - 4c < 0$
 $36 < 4c$ ✓
 $c > 9$

ii) For $x^2 + bx + 20$, this quadratic expression has the same form as $x^2 + bx + c$, and $c > 9$, thus there are no real solutions to $x^2 + bx + 20 = 0$. Also the parabola is upright since $a > 0$. $\therefore x^2 + bx + 20 > 0$ for all x . The graph of $y = x^2 + bx + 20$ is thus completely above the x axis and upright. ✓



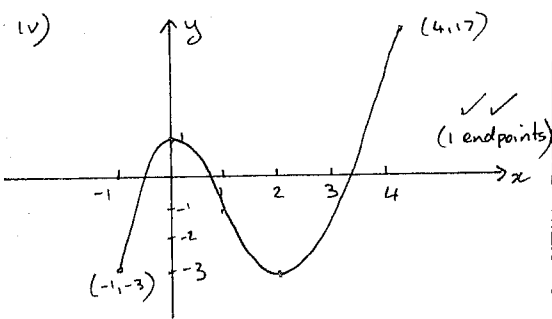
b) $y = x^3 - 3x^2 + 1$
 i) $y' = 3x^2 - 6x$
 $= 3x(x-2)$
 For stat pts $y' = 0$
 ie $3x(x-2) = 0$ ✓
 $x = 0, x = 2$ ✓
 $y'' = 6x - 6$
 when $x = 0$, $y'' = -6$ $\therefore$ max ✓
 when $x = 2$, $y'' = 6$ $\therefore$ min ✓
 y values, when $x = 0$, $y = 1$ ✓
 when $x = 2$, $y = 8 - 12 + 1 = -3$ ✓
 $\therefore (0, 1)$ is a max turning pt
 $(2, -3)$ is a min turning pt

6b) ii) Pts of inflexion, $y'' = 0$
 $6x - 6 = 0$
 $x = 1$ ✓

x	1	1	1
y''	-	0	+

 When $x = 1$, $y = 1 - 3 + 1 = -1$ ✓
 $\therefore (1, -1)$ is a pt of inflexion

iii) Concave down, $y'' < 0$
 $6x - 6 < 0$
 $x < 1$ ✓



$x = -1$, $y = -1 - 3 + 1 = -3$
 $x = 4$, $y = 64 - 48 + 1 = 17$

7 a) $-6 \leq 2x \leq 5 - 2x$
 $-6 \leq 2x$, $2x \leq 5 - 2x$ ✓
 $x \geq -3$, $4x \leq 5$
 $x \leq \frac{5}{4}$ ✓
 $\therefore -3 \leq x \leq \frac{5}{4}$

b) i) $P(D) = \frac{3}{12} = \frac{1}{6}$ (1st draw) ✓
 ii) $P(\text{at least 1 D}) = 1 - P(BB)$ ✓

$= 1 - \frac{5}{6} \times \frac{9}{11}$ ✓
 $= 1 - \frac{15}{22}$
 $= \frac{7}{22}$ ✓
 c) For horizontal pt of inflexion $f'(x) = 0$ and $f''(x) = 0$ ✓
 An example of a function that has a ^{horiz} pt of inflexion is $f(x) = x^3$ ✓
 $C = 2$

d) i) $\frac{dy}{dx} = \frac{x}{x^2 - 4}$
 $y = \int \frac{x}{x^2 - 4} dx$ ✓
 $= \frac{1}{2} \int \frac{2x}{x^2 - 4} dx$
 $= \frac{1}{2} \ln(x^2 - 4) + C$ ✓
 $y = 0$ when $x = 3$
 $\therefore 0 = \frac{1}{2} \ln(9 - 4) + C$
 $0 = \frac{1}{2} \ln 5 + C$
 $C = -\frac{1}{2} \ln 5$ ✓

$\therefore y = \frac{1}{2} \ln(x^2 - 4) - \frac{1}{2} \ln 5$
 ii) $x^2 - 4 > 0$
 $\therefore x^2 > 4$
 ie $x < -2$, $x > 2$ ✓

8 a) $y = \frac{1}{2} \sqrt{4 - x^2}$
 i)

x	-2	-1	0	1	2
y	0	$\frac{\sqrt{3}}{2}$	1	$\frac{\sqrt{3}}{2}$	0

 Area $\doteq \frac{b-a}{6} \{ f(a) + 4f(\frac{a+b}{2}) + f(b) \}$
 Since symmetrical,
 Area $\doteq 2 \times \frac{2-0}{6} \{ f(0) + 4f(1) + f(2) \}$

a) i) Area $\div \frac{2}{3} \left\{ 1 + 4 \times \frac{\sqrt{3}}{2} + 0 \right\}$ ✓
 $= \frac{2}{3} (1 + 2\sqrt{3})$ units² ✓
 $\div 2.976$ units²

ii) $V = \pi \int_{-2}^2 \left(\frac{1}{2} \sqrt{4-x^2} \right)^2 dx$ ✓
 $= 2\pi \int_0^2 \frac{4-x^2}{4} dx$ ✓
 $= \frac{\pi}{2} \int_0^2 (4-x^2) dx$ ✓
 $= \frac{\pi}{2} \left[4x - \frac{x^3}{3} \right]_0^2$ ✓
 $= \frac{\pi}{2} \left(8 - \frac{8}{3} - (0-0) \right)$ ✓
 $= \frac{8\pi}{3}$ units³ ✓

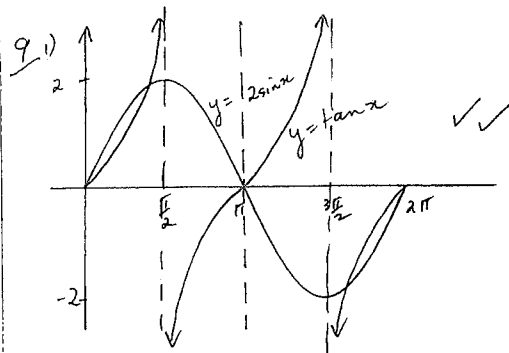
b) $A = 80e^{-0.025t}$

i) $t=10$, $A = 80e^{-0.25}$ ✓
 $\div 62.3g$

ii) Initial amt = 80g
 Half life $\rightarrow 40g$
 $40 = 80e^{-0.025t}$ ✓
 $\frac{1}{2} = e^{-0.025t}$

$\ln \frac{1}{2} = -0.025t$ ✓
 $t = \frac{\ln \frac{1}{2}}{-0.025}$ ✓
 $\div 27.73$ years

iii) $\frac{dA}{dt} = 80 \times -0.025 e^{-0.025t}$ ✓
 $= -2e^{-0.025t}$
 $t=8$ $\frac{dA}{dt} = -2e^{-0.2} \div 1.64g/year$ ✓



ii) There are 5 solutions to $2 \sin x = \tan x$ from the graph ✓

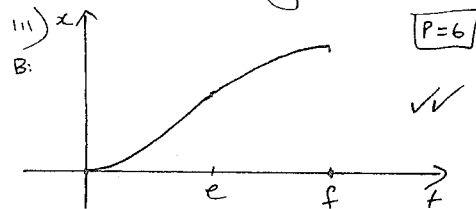
iii) $2 \sin x = \tan x$
 $2 \sin x = \frac{\sin x}{\cos x}$
 $2 \sin x \cos x = \sin x$ ✓
 $2 \sin x \cos x - \sin x = 0$
 $\sin x (2 \cos x - 1) = 0$

iv) $\sin x = 0$, $2 \cos x - 1 = 0$
 $x = 0, \pi, 2\pi$ $\cos x = \frac{1}{2}$ ✓
 $x = \frac{\pi}{3}, \frac{5\pi}{3}$

$\therefore$ Solns are $x = 0, \frac{\pi}{3}, \pi, \frac{5\pi}{3}, 2\pi$ ✓

b) i) A shows the greater accⁿ since gradients of tangents for A are always greater than gradients of tangents for B & $\frac{dv}{dt} = \text{acceleration}$ ✓

ii) Both A & B have the same velocity at $t=e$, however B is accelerating and A is decelerating. ✓



10

a) i) $OR = \pi$

$\therefore QR = e^{-x^2}$

$\therefore \text{Area PQRS} = 2x \times e^{-x^2}$ ✓
 $= 2xe^{-x^2}$

ii) $\frac{dA}{dx} = 2x \times -2xe^{-x^2} + 2e^{-x^2}$ ✓
 $= -4x^2e^{-x^2} + 2e^{-x^2}$
 $= 2e^{-x^2}(1-2x^2)$

For max Area, $\frac{dA}{dx} = 0$

$\therefore 2e^{-x^2}(1-2x^2) = 0$ ✓

$2e^{-x^2} \neq 0$, $1-2x^2 = 0$

$2x^2 = 1$

$x^2 = \frac{1}{2}$

$x = \pm \frac{1}{\sqrt{2}}$ ✓

But initially $x > 0 \therefore x = \frac{1}{\sqrt{2}}$

Check max

$\frac{d^2A}{dx^2} = 2e^{-x^2}(-4x) + 4xe^{-x^2}(-2x^2)$
 $= -8xe^{-x^2} - 4xe^{-x^2} + 8x^3e^{-x^2}$
 $= 4xe^{-x^2}(2x^2-3)$

When $x = \frac{1}{\sqrt{2}}$, $2x^2-3 = 2 \times \frac{1}{2} - 3$

$= -1 < 0$

$\therefore \max(4xe^{-x^2} > 0)$

$\therefore x = \frac{1}{\sqrt{2}}$ gives max area of PQRS ✓

b) i) $f(x) = 2 - \log_e x$

$f'(x) = -\frac{1}{x} < 0$ for all $x > 0$

($\log_e x$ only exists for $x > 0$)

$\therefore f(x)$ is always decreasing

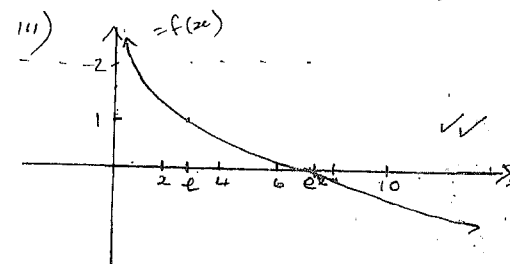
ii) x intercept, $f(x) = 0$

i.e. $0 = 2 - \ln x$ ✓

$\ln x = 2$

$x = e^2$ ✓

$\therefore$ x intercept = $(e^2, 0)$



iv) $A = \int_0^2 f(y) dy$

$y = 2 - \ln x$

$\ln x = 2 - y$

$x = e^{2-y}$

$\therefore A = \int_0^2 e^{2-y} dy$ ✓